

Sophia Sipayboun

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PROFESSIONAL SUMMARY

Bioinformatics graduate student and software engineering graduate with experience in biological data analysis, healthcare analytics, machine learning, and software development. Proficient in Python, R, and SQL, with hands-on experience developing computational tools for genomic and transcriptomic data analysis. Interested in applying computational methods to biological research, precision medicine, and healthcare.

EDUCATION

Northeastern University, M.S. in Bioinformatics

Expected May 2028

St. Petersburg College, B.S. in Software Engineering, GPA: 3.7 / 4.0

Aug 2022 - May 2026

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, C++, JavaScript

Bioinformatics & Data Analysis: Biopython, Genomic & Transcriptomic Data Analysis, Gene Expression Analysis, PCA/UMAP, Exploratory Data Analysis, Data Visualization

Machine Learning: Classification Models, Model Evaluation, Data Preprocessing

Databases & Tools: MySQL, MongoDB, Flask, Git, Cloud Computing

Additional: Data Pipelines, Web Scraping, Quality Assurance

WORK EXPERIENCES

Healthcare Data Analyst Extern, TruBridge

Remote, FL

Extern

Feb 2026 - May 2026

- Conducted exploratory data analysis on publicly available healthcare datasets, examining infection trends through variables like case counts, demographics, and time patterns to uncover meaningful insights.
- Leveraged structured research and analytical tools to guide data discovery, generate research prompts, and support the creation of visualizations, dashboards, and reports on infection-related public health trends.
- Produced synthesis reports and presentations covering key areas including infection trends, data limitations, population insights, public health implications, and recommendations for stakeholder action.

Software Developer Intern

Remote, FL

Jetzy

Aug 2022 – May 2023

- Developed a web-based visualization platform to analyze user behavior and engagement data.
- Built Python/Flask workflows to extract and process data from SQL databases for downstream analysis.
- Applied machine learning classification and web scraping techniques to categorize online content and analyze user engagement.

PROJECTS

GeneScope | Python, Flask, Pandas, scikit-learn, UMAP, GSEAPy

Apr 2026

- Developed a full-stack bioinformatics platform for exploration and visualization of transcriptomic gene expression data
- Implemented preprocessing and quality-control workflows alongside gene expression heatmaps, PCA/UMAP dimensionality reduction, differential expression previews, and pathway enrichment.
- Designed an interactive interface for exploring large RNA-seq datasets and translating computational analyses into interpretable visualizations

Gene Sequence Analyzer | Python, Biopython, Flask, JavaScript

May 2026

- Developed an interactive bioinformatics application for DNA sequence analysis, mutation detection, ORF discovery, and protein translation.
- Implemented sequence-processing workflows with Biopython and integrated analyses into a Flask-based web application.
- bbbb

ADDITIONAL TRAINING & ORGANIZATIONS

Stanford Data Ocean Bioinformatics Bootcamp (via Conectado)

Member

Dec 2023 – Jan 2024

- Completed career readiness program in collaboration with Stanford Data Ocean Bioinformatics Bootcamp
- Earned a certificate in Fundamentals of Data Science in Precision Medicine and Cloud Computing

Rewriting the Code

Member

May 2022 – Present

- Engaged in mentorship and networking opportunities with women in technology, strengthening professional development and career readiness
- Participated in technical workshops and career panels focused on coding, leadership, and industry best practices